

Don't Stop Retrievin' [*]: Ensuring Version Integrity in Publications and Metadata

Versioning via the Railway Coupling Method (aka the “Treintjesmodel”) for Asynchronously Submitted Publications and Metadata

ABSTRACT

This poster shows the specialized versioning workflow for e-books that was developed by KB National Library of the Netherlands (KBNL) in 2025. What makes this workflow unique and special, is that

- it marks the first implementation of versions in KBNL's e-Depot,
- combined with the way these e-books and associated metadata are sent to KBNL: separately and asynchronously.

[*] Journey - Don't Stop Believin' (Official Audio) :



- With this solution, users will not miss a single version of the publication or the metadata. End-users can retrieve all versions at any time. They don't have to stop believin' that they might be missing something [FAIR] as long as they don't stop retrievin' (to fulfill the R in the FAIR principle).
- Furthermore, this implementation eliminates the storage of duplicate publications and metadata!

KBNL VERSION POLICIES

- **Full version management:** All modifications to digital objects, metadata, and content are stored as new versions to ensure version integrity.
- **Relationship preservation:** Connections between versions are kept transparent, making it possible to manage and track the object's chain of custody.
- **Event metadata:** Details of every action performed on the digital objects are recorded to verify correct processing and provenance.

STANDARDS & BEST PRACTICES

- **OAIS** distinguishes AIP Versions (triggered by migrations performed by the Archive) and AIP Editions (triggered by update requests from the Producer)
- **CTS** references versioning data and metadata in (R07) Provenance and authenticity
- **FAIR** principles F1. (Meta)data are assigned a globally unique and persistent identifier / R1.2. (Meta)data are associated with detailed provenance
- **EOSC EDEN** identifies AIP Versioning as one of the Core Preservation Processes - CPP-021 AIP Versioning

KBNL VERSION REQUIREMENTS & DECISIONS

Publications

- **Version Creation:** All changes made to a publication by the Producer result in a new IE/AIP. Each IE/AIP is assigned its own unique PID (URN:NBN).
- **Version Grouping via SOURCE ID:** The SOURCE ID field determines which IEs/AIPs are versions of one another. For CB e-books, the ISBN is used as the SOURCE ID to group all related versions.
- **Version Sequencing:** The CAPTURE date of the publication determines the chronological sequence of the versions.
- **The "Treintjesmodel" (Chaining):** A new version of a publication references the PID (URN:NBN) of its immediate predecessor in the ISVERSIONOF field. Versions are linked sequentially like train carriages; there is no backward reference from an older version to a newer one. (For the initial version, the ISVERSIONOF field remains empty).

Metadata

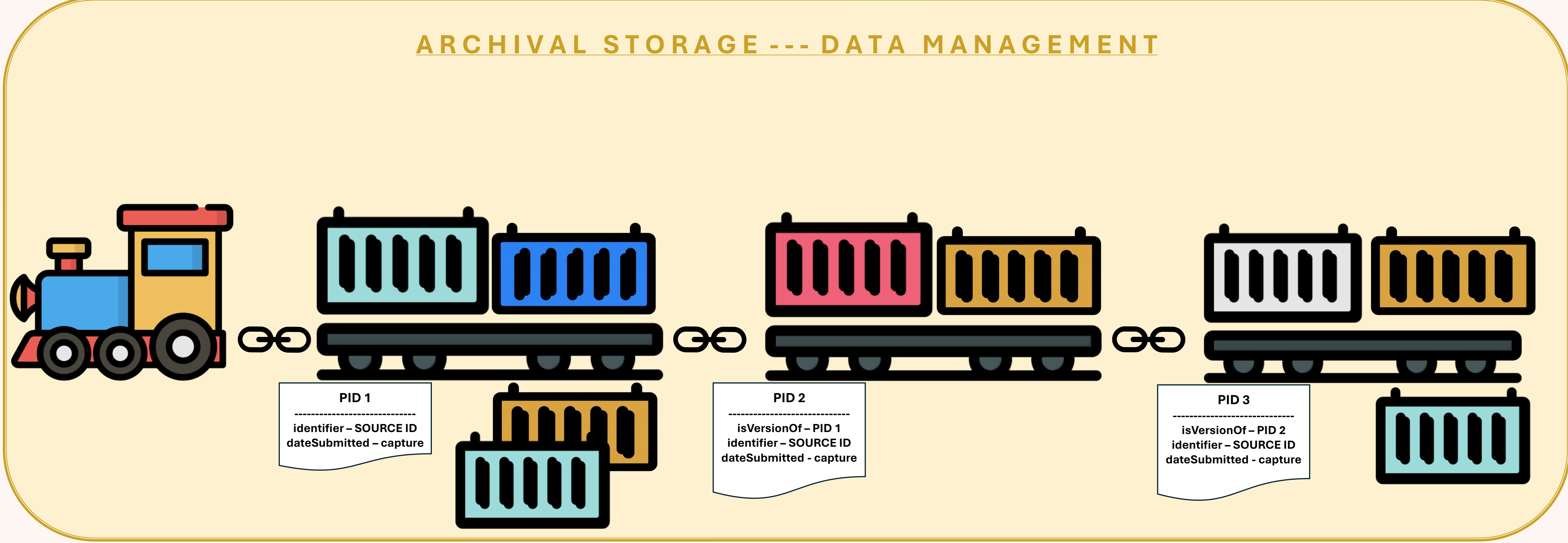
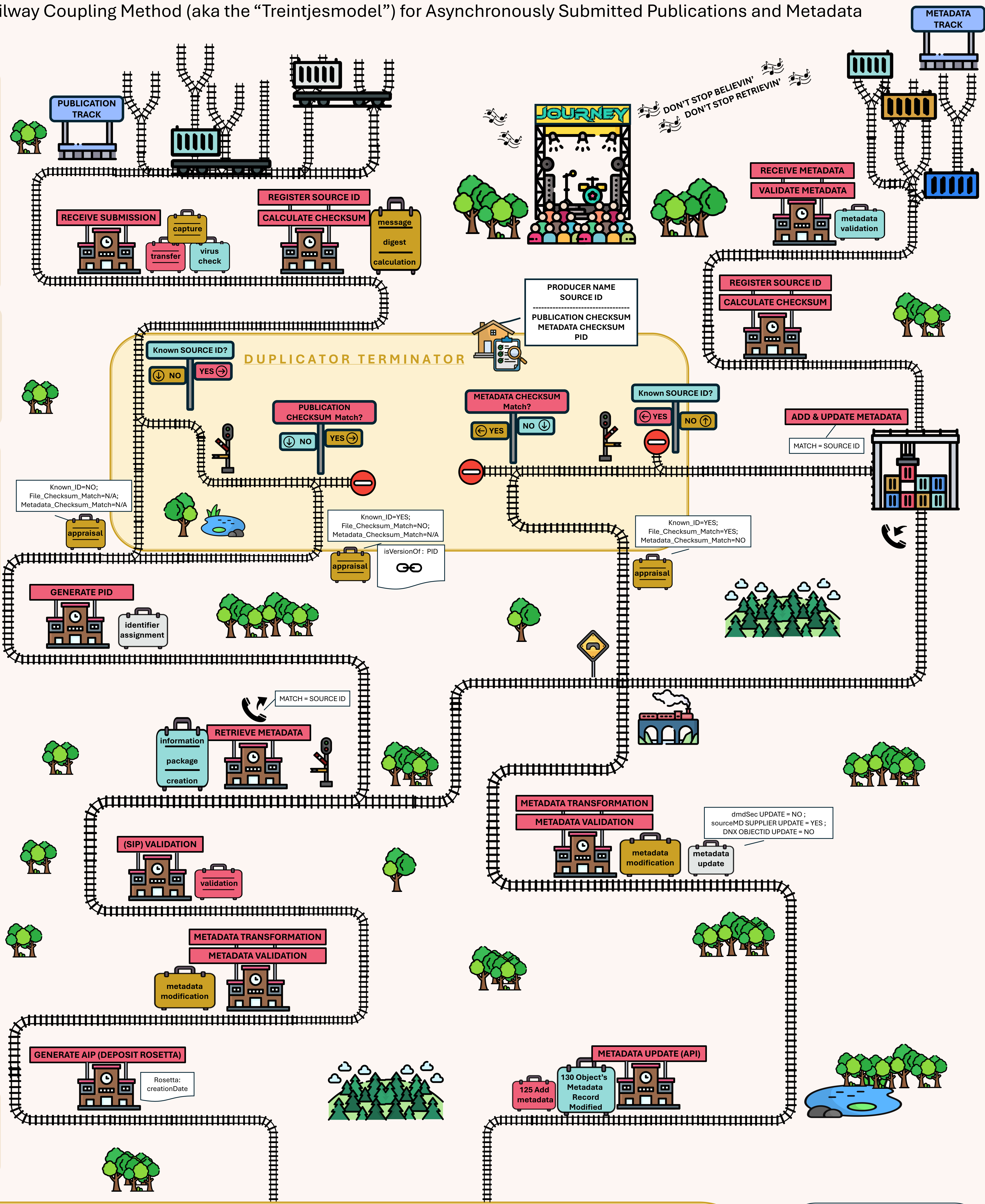
- **Supplied metadata storage:** Supplied metadata is not preserved as a file, but is stored as embedded source metadata.
- **Metadata Updates:** Metadata updates are processed by Rosetta and will generate a new version of the existing IE/AIP, in accordance with standard Rosetta functionality.
- **Target Publication Version for Updates:** To minimize version proliferation, metadata updates are exclusively applied to the most recent IE/AIP in the publication version chain. This is based on the business rule that metadata updates always apply to the latest version of a publication.

FUTURE WORK

- **Synchronous Versioning Workflow:** Development of a versioning workflow for publications and metadata that are received synchronously.
- **User Access and Retrieval:** Implementation of a delivery workflow that enables end-users to both view and request all distinct versions of the publications and metadata.

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- We would like to thank all KBNL colleagues who contributed in any way to the realization of the version control workflow for e-books
- This poster has been designed using resources from Flaticon.com and Magnific.com
- The railroad tracks have been made with the help of ChatGPT



LEGEND

- Flow
- Process
- Decision
- Process : Duplicator Terminator
- Temporary Metadata Storage
- Provenance metadata : Events
- Publication / content
- Metadata
- Coupling / link
- Delay

